

AI-DRIVEN MAINTENANCE REPORT

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1. Technical Sensor Data

Parameter	Measured Value
Air Temperature	300.0 °F
Process Temperature	340.0 °F
Rotational Speed	1200 RPM
Torque	65.0 Units
Tool Wear	210 %

2. AI Agent Analysis

Agent 1 (Initial Analysis):

Torque > 60 and Tool wear > 200 indicate high risk.

Agent 2 (Verification):

Torque >60 and Tool wear >200 trigger high risk rules.

3. Engineering Analysis & Recommendations

Summary:

REQUIRED MAINTENANCE

Analysis:

The sensor data indicates that the system is experiencing high-risk conditions. Specifically, the torque value of 65.0 exceeds the safety threshold of 60, indicating "high torque". Additionally, the tool wear value of 210 exceeds the safety threshold of 200, indicating "tool wear high". Furthermore, the air temperature of 300.0 and process temperature of 340.0 indicate an abnormal temperature difference of 40.0, exceeding the safety threshold of 25.

Validation:

The sensor values exceed the established thresholds as follows:

- * Torque > 60: Yes (65.0 > 60)
- * Tool Wear > 200: Yes (210 > 200)
- * Temperature Difference > 25: Yes (40.0 > 25)

Recommendation:

1. IMMEDIATELY reduce rotational speed to minimize further tool wear and torque buildup.
2. SCHEDULED maintenance is required to inspect and replace the worn-out tools to prevent damage to other system components.
3. PERFORM an urgent temperature control adjustment to bring the process temperature within the acceptable range, ensuring safe operation of the system.

4. Conclusion

Final Decision: REQUIRED MAINTENANCE

Decision Source: RULE-BASED SYSTEM TRIGGERED